

Myntra Database Management System

Introduction:

Myntra Database Management System is a database-driven project designed to manage and streamline the core operations of an online fashion and lifestyle e-commerce platform, similar to Myntra. The system handles data related to customers, products, orders, payments, inventory, and sellers, ensuring efficient storage, retrieval, and management of large volumes of transactional and user data. This project focuses on designing a robust relational database structure that supports the day-to-day functioning of an e-commerce platform while maintaining data integrity, consistency, and security.

Problem Statement:

E-commerce platforms like Myntra deal with massive amounts of data every day, including customer details, product catalogs, order histories, payment transactions, and inventory records. Managing this data manually or through poorly structured systems leads to redundancy, inconsistency, slow query performance, and difficulty in scaling operations. There is a need for a well-structured, normalized database system that can efficiently store, manage, and retrieve data while supporting the various functionalities required by an online shopping platform, such as order processing, inventory tracking, and customer management.

Objectives:

- 1.To design a normalized relational database for an e-commerce platform.
- 2.To efficiently manage customer, product, order, and payment data.
- 3.To ensure data integrity, consistency, and minimal redundancy.
- 4.To support fast and accurate data retrieval for business operations.

Key Objectives:

1. Design entity-relationship (ER) diagrams representing the database structure.
2. Implement tables for customers, products, categories, orders, order items, payments, sellers, and inventory.
- 3.Establish proper primary key and foreign key relationships between tables.
- 4.Apply normalization techniques to eliminate data redundancy.

5. Write SQL queries to perform CRUD (Create, Read, Update, Delete) operations.
6. Ensure data security and integrity through constraints and validations.

Motivation:

With the rapid growth of online shopping platforms in India, companies like Myntra process millions of transactions and manage vast product catalogs on a daily basis. This project is motivated by the desire to understand how large-scale e-commerce platforms structure their backend databases to handle such complexity efficiently. Building a scaled-down version of such a system provides practical, hands-on experience in database design, normalization, and query optimization skills that are highly relevant in real-world software development and data management roles.

Business Requirements:

1. The system must store and manage customer information including name, contact details, and addresses.
2. The system must maintain a product catalog with details such as category, price, brand, size, and stock availability.
3. The system must record order details including order date, order status, and items purchased.
4. The system must track payment information including payment mode and transaction status.
5. The system must maintain seller/vendor information for products listed on the platform.
6. The system must support inventory management to track stock levels for each product.
7. The system must allow generation of reports such as sales summaries and order histories.

Stakeholders:

- 1. Customers** – Users who browse and purchase products on the platform.
- 2. Sellers/Vendors** – Businesses or individuals who list products for sale.
- 3. Administrators** – Personnel who manage the platform's backend operations, including product listings, orders, and user accounts.
- 4. Delivery Partners** – Entities responsible for order fulfillment and shipping.
- 5. Development Team** – Individuals responsible for designing, building, and maintaining the database system.

6. Business Management – Decision-makers who rely on data insights for strategic planning.

Functional Requirements:

1. The system shall allow registration and login for customers and sellers.
2. The system shall allow customers to browse products by category, brand, and price range.
3. The system shall allow customers to add products to a cart and place orders.
4. The system shall allow sellers to add, update, and remove product listings.
5. The system shall record and update order status (e.g., placed, shipped, delivered, cancelled).
6. The system shall process and store payment transaction details.
7. The system shall update inventory levels automatically after each purchase.
8. The system shall allow administrators to view and manage all orders, products, and users.
9. The system shall support search functionality for products.

Non-Functional Requirements:

- 1. Performance** – The database should support quick query execution even with large datasets.
- 2. Scalability** – The system should be able to handle an increasing number of users, products, and transactions.
- 3. Reliability** – The system should ensure accurate and consistent data storage without data loss.
- 4. Security** – Customer and payment data must be protected through proper access controls and validation.
- 5. Maintainability** – The database schema should be easy to update and extend as requirements evolve.
- 6. Availability** – The system should be designed to minimize downtime and ensure continuous access to data.
- 7. Usability** – The database structure should support easy integration with front-end applications for a smooth user experience.